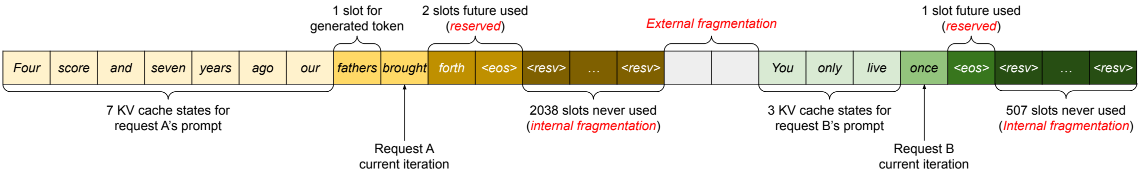

re 3. KV cache memory management in existing systems. Three types of memory wastes – reserved, internal fragmenta
xternal fragmentation – exist that prevent other requests from fitting into the memory. The token in each memory
esents its KV cache. Note the same tokens can have different KV cache when at different positions.

Memory Challenges in LLM Serving

ough fine-grained batching reduces the waste of com-
ng and enables requests to be batched in a more flexible
the number of requests that can be batched together is
constrained by GPU memory capacity, particularly the
e allocated to store the KV cache. In other words, the
ing system's throughput is *memory-bound*. Overcom-
his memory-bound requires addressing the following
enges in the memory management:

§6.3) of their KV cache, and the sharing pattern evolve
the decoding process advances.

Scheduling for unknown input & output lengths.
requests to an LLM service exhibit variability in their i
and output lengths. This requires the memory managen
system to accommodate a wide range of prompt length
addition, as the output length of a request grows at deco
the memory required for its KV cache also expands and
exhaust available memory for incoming requests or ong
generation for existing prompts. The system needs to m